

User Manual: DA Graphical User Interface for ISO

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**Centre for Military Airworthiness & Certification (CEMILAC)
Defence Research and Development Organisation (DRDO)**

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Revision History

Sl. No.	Version	Date	Changes
1.	0.1	June 2023	Original
2.	0.2	July 2023	Updated as per CAIR observation on design details
3.	0.3	September 2023	Updated after the integration Testing
4.	0.4	January 2026	Updated user manual with New GUI Screenshots and documentation standardization

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Acronyms	
CEMILAC	Centre for Military Airworthiness and Certification
CSS	CAIR Security Server
CWS	CEMILAC Web Server
DA	Design Agency
IBPS	Business Processing System
ISO	International Standardization Organization
RCMA	Regional Centre for Military Airworthiness

1 Introduction

The DA-ISO Graphical User Interface (GUI) is designed to provide the option of uploading the document, project registration form from Design Agency (DA) to CEMILAC Web Server. The interface will also allow the DA to download the observations/comment sent by Certification Agency.

This document describes the DA-ISO GUI and the various actions to be performed to interface with CSS-I.

2 DA-ISO GUI

2.1 Login Page

After placing the security dongle and booting the ISO Image, the home page of DA-ISO web interface will open. The webpage can also be opened by placing the URL **localhost:9002/iso/** in browser as shown in Figure-2. User authentication to the DA-ISO web interface is enforced exclusively through a DA security token. The system validates the token during login to ensure authorized access. The DA security token should be case sensitive.



Figure 1 Login page

2.1.1 Connection Failed

In case the CAIR Security Server (CSS) is not reachable during the login; the warning message as shown in Figure-3 will be displayed.

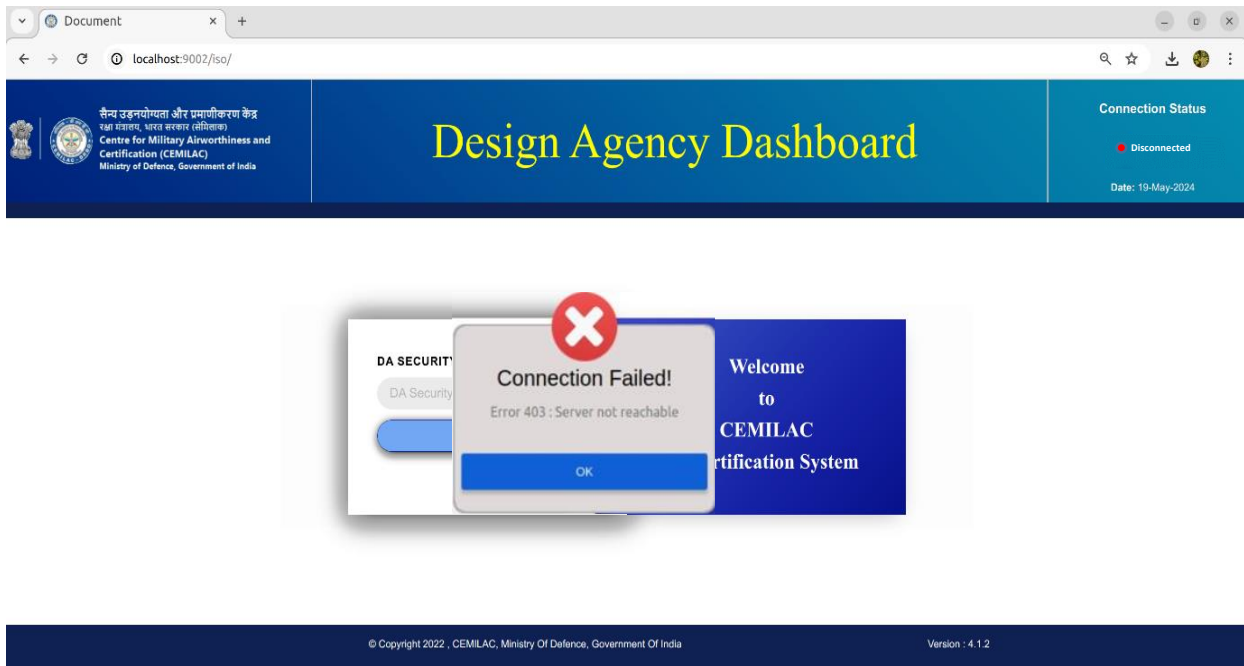


Figure 2 Connection failed during login

2.1.2 Download Failed

The very first action DA-ISO webpage will carry out is to download of **Static_<DAID>.tar** file from CSS. In case static file is not present at CSS; the warning message as shown in Figure-4 will be displayed.

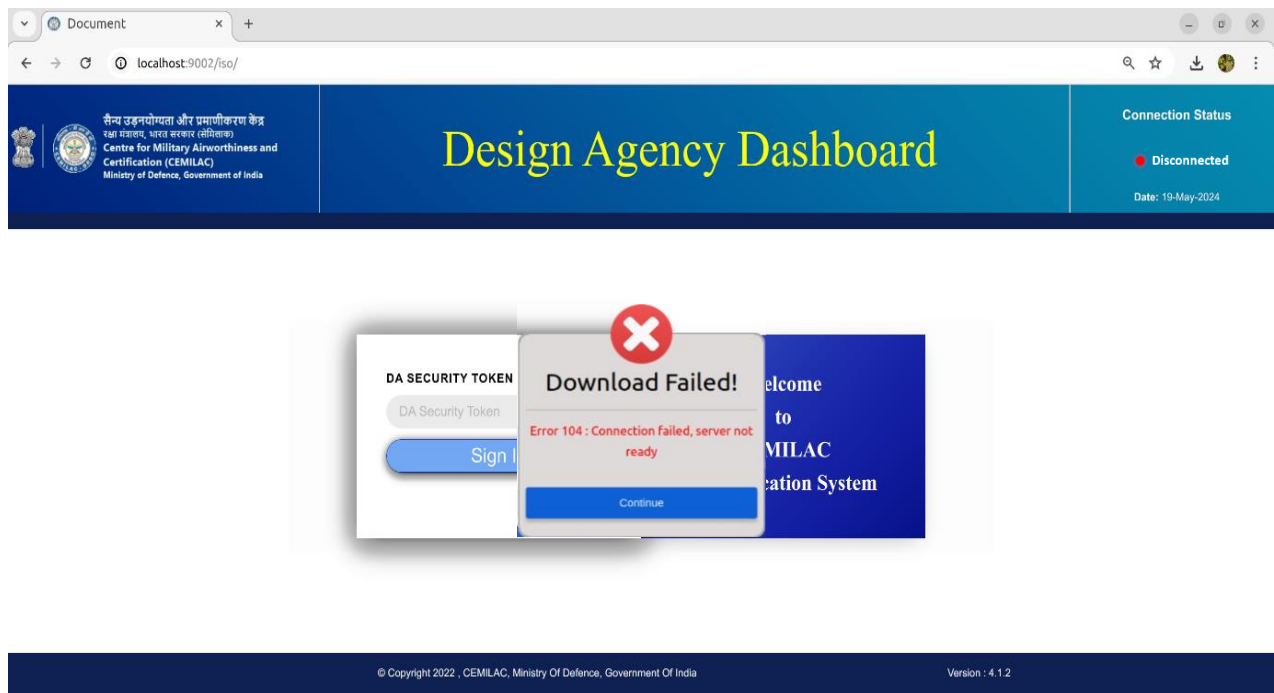


Figure 3 Static file download failed

2.1.3 Login Failed

The failure of static file download or corruption of static file content will lead to login failed as login credentials are saved in static file.



Figure 4 Invalid static file or corrupt static file

2.2 Landing Page after Login: Registered Projects

After successful login, the DA-ISO application will download the **Dynamic_<DA-ID>.tar** file. This file contains the information about the previous transaction status and observations given by CEMILAC.

2.2.1 Malicious transaction alert

In case the previous documents/artefact submitted by Design Agency had virus/malware, the content will be dropped by CSS and an alert message as shown in Figure6 will be displayed.

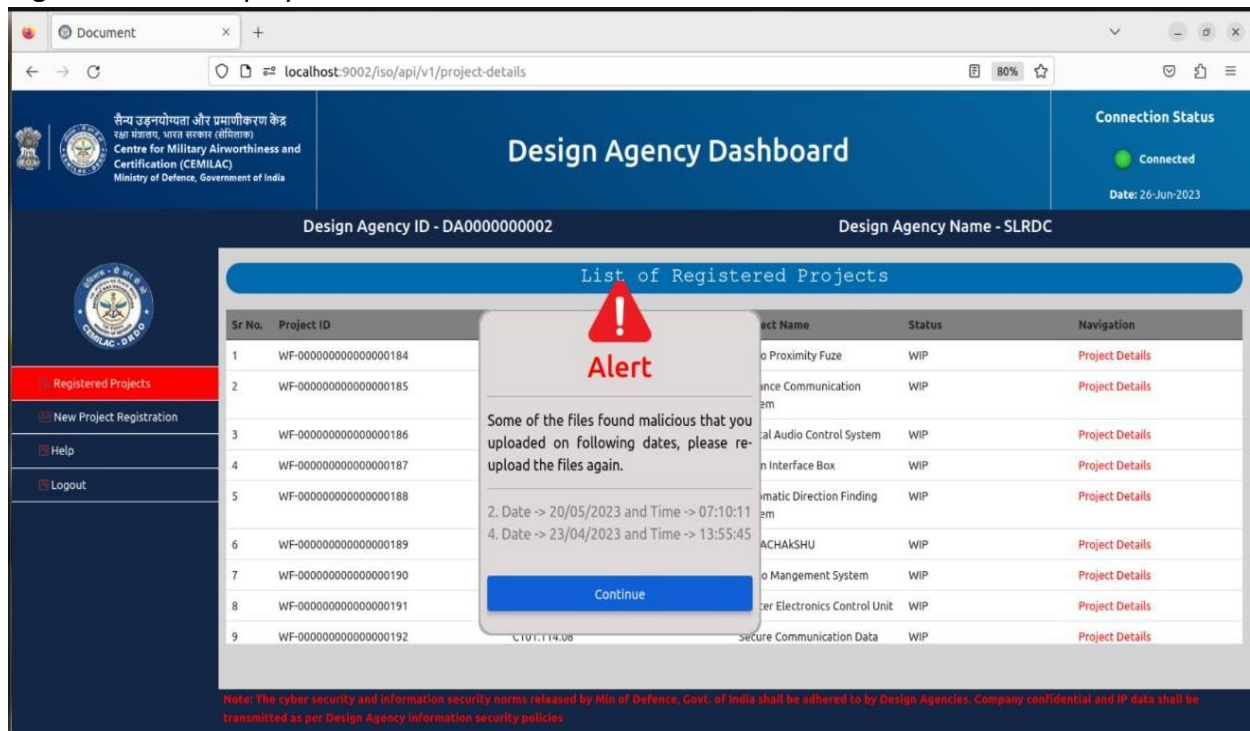


Figure 5 Alert: Last transaction malicious

2.2.2 Observations alert

In case the observations are received from CEMILAC, the Design Agency will be alerted to save the observations prior to logout as shown in Figure-7.

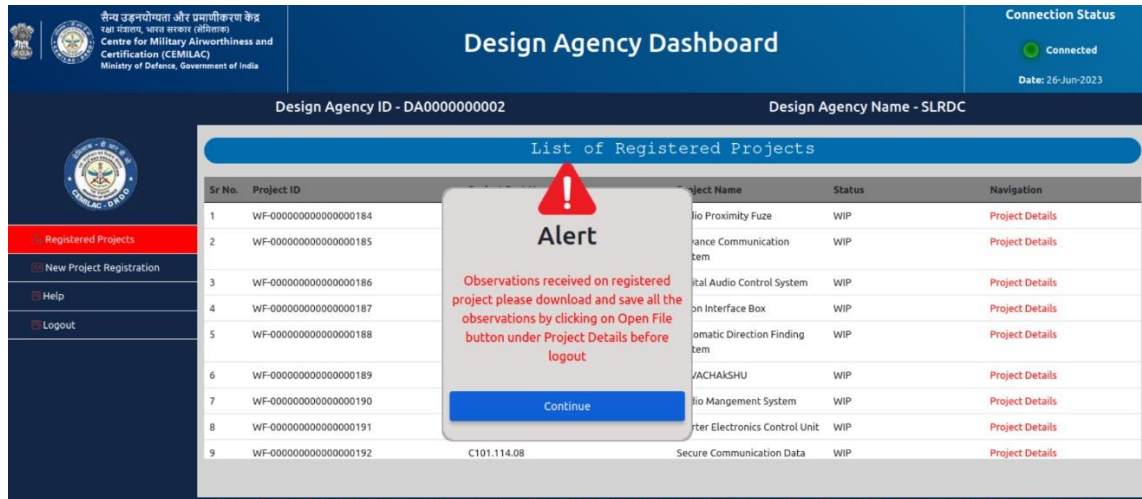


Figure 6 Alert: The DA shall save the observation prior to logout

2.2.3 Registered Project Details

After alert messages the list of Registered Projects with Design Agency will be displayed as shown in Figure-8. The DA can navigate to Project Details by clicking on **Project Details**.

सैन्य उड़ानयोग्यता और प्रमाणीकरण केंद्र रक्षा मंत्रालय, भारत सरकार (संयुक्त) Centre for Military Airworthiness and Certification (CEMILAC) Ministry of Defence, Government of India		Design Agency Dashboard			Connection Status Connected Date: 26-Jun-2023
Design Agency ID - DA0000000002 Design Agency Name - SLRDC		List of Registered Projects			
Sr No.	Project ID	Project Part Number	Project Name	Status	Navigation
1	WF-000000000000000184	C101.114.00	Radio Proximity Fuze	WIP	Project Details
2	WF-000000000000000185	C101.114.01	Advance Communication System	WIP	Project Details
3	WF-000000000000000186	C101.114.02	Digital Audio Control System	WIP	Project Details
4	WF-000000000000000187	C101.114.03	Pylon Interface Box	WIP	Project Details
5	WF-000000000000000188	C101.114.04	Automatic Direction Finding System	WIP	Project Details
6	WF-000000000000000189	C101.114.05	NAVACHAKSHU	WIP	Project Details
7	WF-000000000000000190	C101.114.06	Audio Mangement System	WIP	Project Details
8	WF-000000000000000191	C101.114.07	Starter Electronics Control Unit	WIP	Project Details
9	WF-000000000000000192	C101.114.08	Secure Communication Data	WIP	Project Details

Note: The cyber security and information security norms released by Min of Defence, Govt. of India shall be adhered to by Design Agencies. Company confidential and IP data shall be transmitted as per Design Agency information security policies

Figure 7 List of Registered Projects

2.2.4 Registered Project Details

After navigating to Project Details by clicking on **Project Details** the details of the documents to be submitted and their status is presented as per the Applicability Matrix. The observations from CEMILAC can be downloaded, the updated/new documents can be uploaded as shown in Figure-9.

Design Agency Dashboard

Connection Status: ● Connected
Date: 26-Jun-2023

Design Agency ID - DA0000000002 Design Agency Name - SLRDC

Applicability Matrix : Document Status

Project ID - WF-00000000000000184

Sr No.	Document Name	Status	Action	CEMILAC Observations	Remarks
1	Functional_Hazard_Analysis	Re-Upload	Browse... No file selected.	Upload	Open file Upload the document after incorporating the corrections
2	FMECA	Accepted	Browse... No file selected.	Upload	Open file The hardcopy may be submitted for signature
3	Reliability_Analysis	Pending	Browse... No file selected.	Upload	Open file
4	Preliminary_Design_Document	Pending	Browse... No file selected.	Upload	Open file
5	Technical_Specification	Pending	Browse... No file selected.	Upload	Open file
6	Functional_Test_Plan	Pending	Browse... No file selected.	Upload	Open file
7	SOFT	Pending	Browse... No file selected.	Upload	Open file

Note: The cyber security and information security norms released by Min of Defence, Govt. of India shall be adhered to by Design Agencies. Company confidential and IP data shall be transmitted as per Design Agency information security policies

Figure 8 Registered Project Details

2.2.5 Document Upload

In order to upload the document to CSS, the document may be browsed and selected for upload as shown in Figure-10. In case the size of document/file selected for upload is greater than 500MB. The warning message as shown in the Figure-11 will be displayed and the document/file will get unselected. The upload is permitted for against the previously uploaded document and the first pending document.

Design Agency Dashboard

Connection Status: ● Connected
Date: 27-Jun-2023

Design Agency ID - DA0000000002 Design Agency Name - SLRDC

File Upload

Cancel Select

Recent Home Documents Downloads Music Pictures Videos

Open files read-only All Files

Name	Size	Type	Modified
APPLICATION_FOR_HW_CLEARANCE_DACS.pdf	324.3 kB	Document	9 Apr
CODE_VHDL.txt	12.3 kB	Text	9 Apr
DYNAMIC_TEST_REPORT_DACS.pdf	324.3 kB	Document	9 Apr
FMECA_Large.pdf	158.3 MB	Document	1 Jan 2013
HARDWARE_CONFIGURATION_MANAGEMENT_PLAN_D...	324.1 kB	Document	9 Apr
HARDWARE_DESIGN_DOCUMENT_DACS.pdf	324.3 kB	Document	9 Apr
HARDWARE_VERIFICATION_AND_VALIDATION_PLAN_...	323.9 kB	Document	9 Apr
HW_REQUIREMENT_DESCRIPTION.pdf	324.3 kB	Document	9 Apr
HW_VERIFICATION_REPORT_DACS.pdf	324.1 kB	Document	9 Apr
HW_VERIFICATION_CASES_AND_PROCEDURE_DACS.pdf	324.4 kB	Document	9 Apr

Note: The cyber security and information security norms released by Min of Defence, Govt. of India shall be adhered to by Design Agencies. Company confidential and data shall be transmitted as per Design Agency information security policies

Figure 9 Browsing the document for Upload

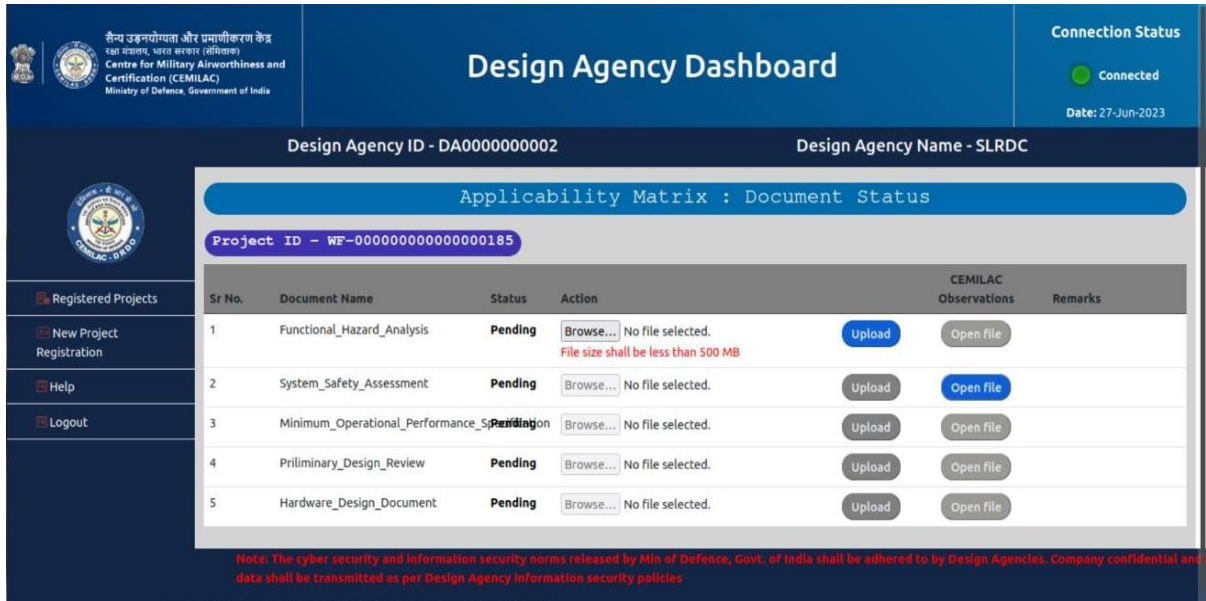


Figure 10 Warning for file size greater than 500MB

If size of document/file selected is less than 500MB, the file will be uploaded to CSS as shown in Figure-12 , 13 & 14.

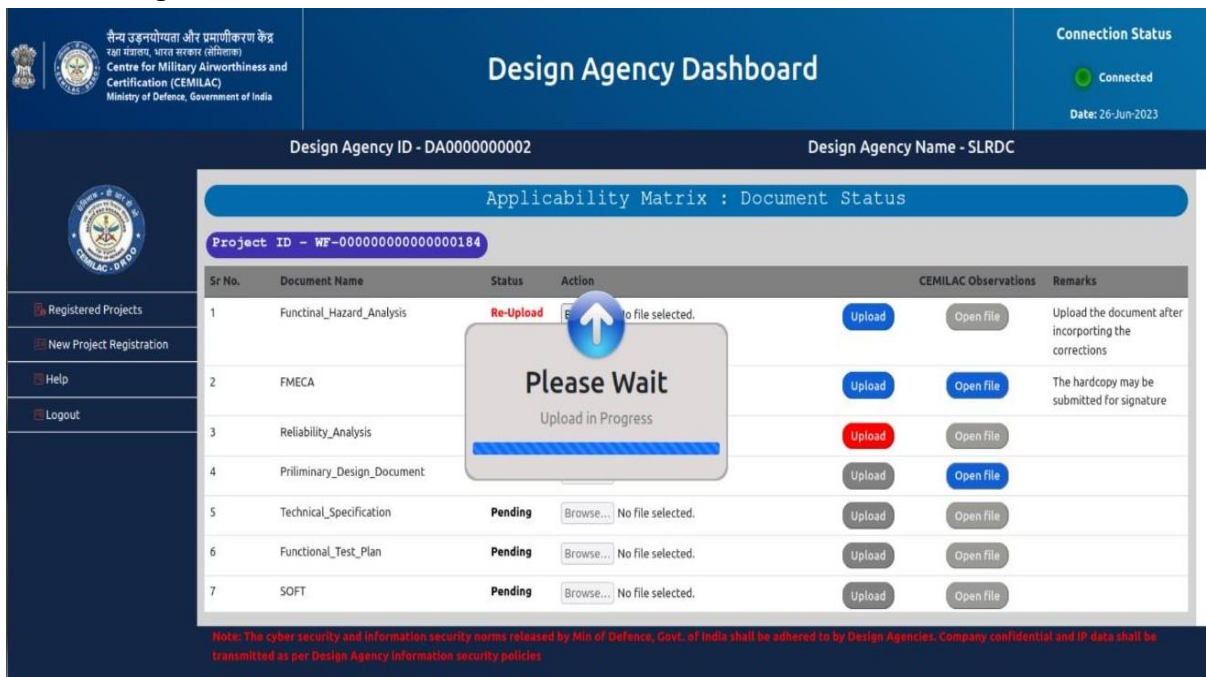
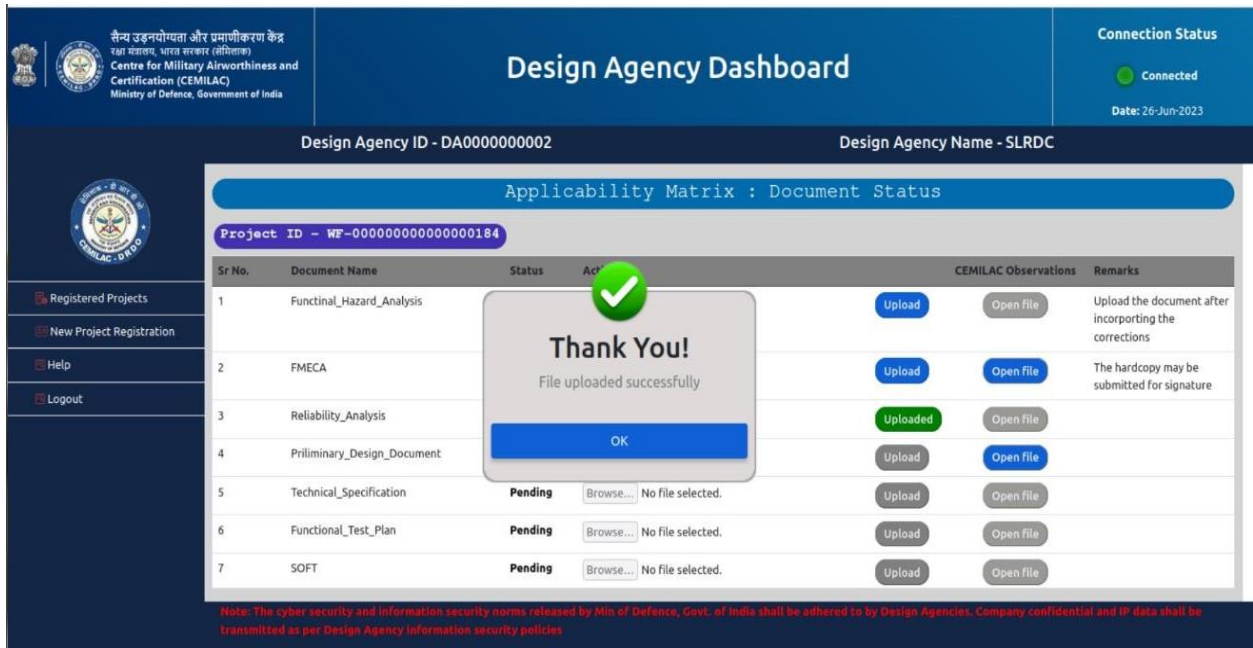


Figure 11 Uploading the file to CSS



Design Agency Dashboard

Connection Status: ● Connected
Date: 26-Jun-2023

Design Agency ID - DA0000000002 Design Agency Name - SLRDC

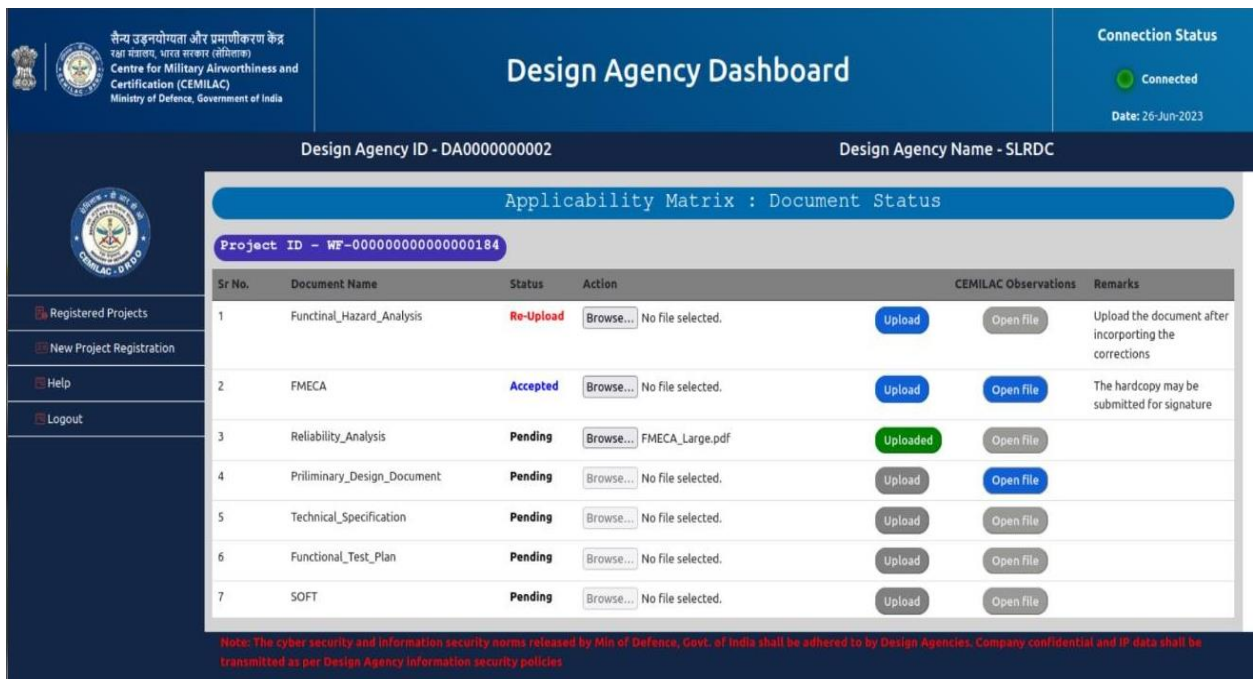
Applicability Matrix : Document Status

Project ID - WF-0000000000000000184

Sr No.	Document Name	Status	Action	CEMILAC Observations	Remarks
1	Functional_Hazard_Analysis	✔	Upload	Open File	Upload the document after incorporating the corrections
2	FMECA	Accepted	Upload	Open File	The hardcopy may be submitted for signature
3	Reliability_Analysis	Pending	Upload	Open File	
4	Preliminary_Design_Document	Pending	Upload	Open File	
5	Technical_Specification	Pending	Upload	Open File	
6	Functional_Test_Plan	Pending	Upload	Open File	
7	SOFT	Pending	Upload	Open File	

Note: The cyber security and information security norms released by Min of Defence, Govt. of India shall be adhered to by Design Agencies. Company confidential and IP data shall be transmitted as per Design Agency information security policies

Figure 12 Successful Upload of the file



Design Agency Dashboard

Connection Status: ● Connected
Date: 26-Jun-2023

Design Agency ID - DA0000000002 Design Agency Name - SLRDC

Applicability Matrix : Document Status

Project ID - WF-0000000000000000184

Sr No.	Document Name	Status	Action	CEMILAC Observations	Remarks
1	Functional_Hazard_Analysis	Re-Upload	Browse... No file selected.	Upload	Open File
2	FMECA	Accepted	Browse... No file selected.	Upload	Open File
3	Reliability_Analysis	Pending	Browse... FMECA_Large.pdf	Uploaded	Open File
4	Preliminary_Design_Document	Pending	Browse... No file selected.	Upload	Open File
5	Technical_Specification	Pending	Browse... No file selected.	Upload	Open File
6	Functional_Test_Plan	Pending	Browse... No file selected.	Upload	Open File
7	SOFT	Pending	Browse... No file selected.	Upload	Open File

Note: The cyber security and information security norms released by Min of Defence, Govt. of India shall be adhered to by Design Agencies. Company confidential and IP data shall be transmitted as per Design Agency information security policies

Figure 13 Project details after upload of the file

2.2.6 Document / Observations Download

In order to download the observations given by CEMILAC, the **Open file** tab may be clicked. This will open the location of the observation file as shown in Figure-15, which need to be saved in the external media.

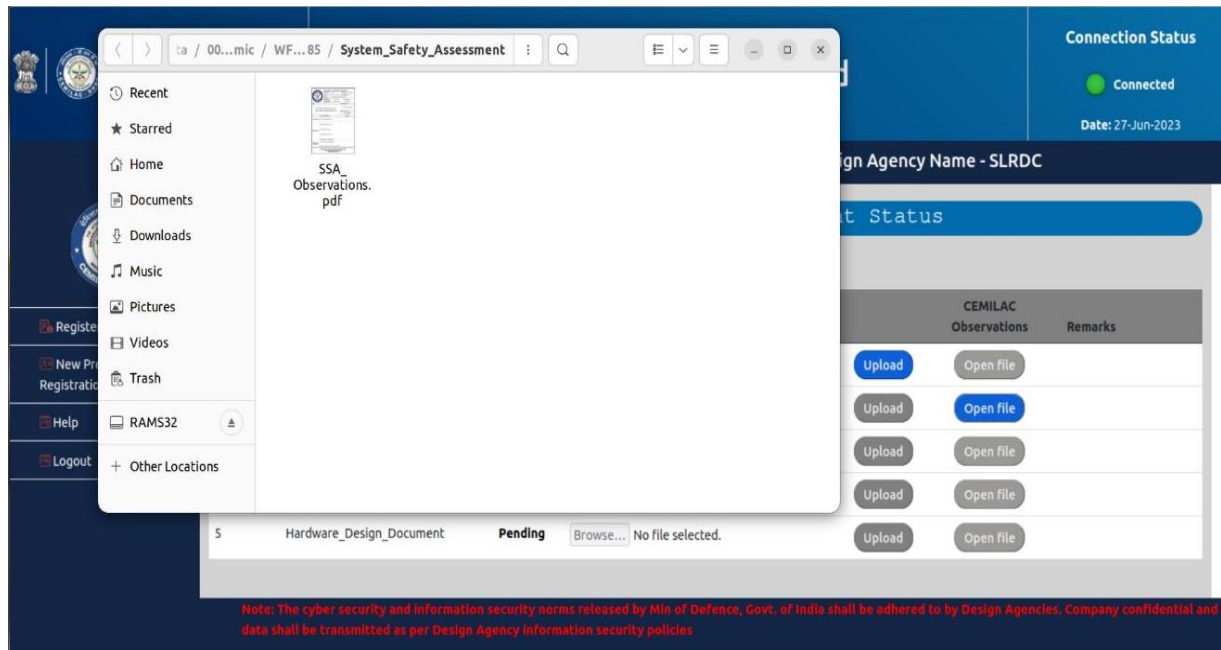


Figure 14 Document / Observation save option

2.3 New Project Registration

2.3.1 Filling the Form

In case Design Agency wish to registered new project for certification the **New Project Registration** shall be clicked. The Project Registration form as shown in Figure 16, 17 & 18 shall be filled and submitted.

Figure 15 Project Registration Form: Contact Information

The screenshot shows the 'Contact Information' tab of the Project Registration Form. The left sidebar contains links for 'Registered Projects', 'New Project Registration' (highlighted), 'Help', and 'Logout'. The main content area has three tabs: 'Contact Information' (active), 'Project Details', and 'View Application'. The 'Contact Information' tab includes a date 'Date: 26-06-2023' and several input fields: 'DOA Details' with a 'DD/MM/YYYY' date picker and 'Select DOA Validity' dropdown; 'DOA Scope' text area; 'POA Details' with a 'DD/MM/YYYY' date picker and 'Select POA Validity' dropdown; 'POA Scope' text area; 'Project Name' text field; a text area for 'Provide details about project' with a '255 Characters Remaining' indicator; 'Project Part Number' text field; and 'Project IMTAR Sub-Part' dropdown. The 'Product Breakdown Structure' section has a 'Choose File (Max Size: 25MB)' button and a 'Browse' button.

Figure 16 Project Registration Form: Project Details

The screenshot shows the 'Project Details' tab of the Project Registration Form. The left sidebar is the same as in Figure 15. The main content area has three tabs: 'Contact Information', 'Project Details' (active), and 'View Application'. The 'Project Details' tab includes a 'POA Scope' text area, a 'System Description' section with 'Project Name' text field and 'Provide details about project' text area (with '255 Characters Remaining' indicator), 'Project Part Number' text field, and 'Project IMTAR Sub-Part' dropdown. The 'Product Breakdown Structure' section has a 'Browse...' button, 'No file selected.' text, and 'Choose File (Max Size: 500MB)' button. The 'Applicant's Declaration' section has three checkboxes: 'I declare that I am authorized by my organization to submit this application to CEMILAC and that all information provided in this application form is correct and complete.', 'I acknowledge that I have read and understood the IMTAR-21.', and 'I understand that the submission of the application does not entitle certification coverage by CEMILAC.'. At the bottom are 'Previous', 'Cancel', 'Reset', and 'Next' buttons.

Figure 17 Project Registration Form: PBS and declaration

2.3.2 Submitting the Form

On completion of the form the filled details will be displayed as shown in the Figure-19. The form can be submitted after verification of the details as shown in Figure-20.

Registered Projects
New Project Registration
Help
Logout

Applicant Reference
Applicant Reference: DA_000000000226062023215714

Contact Person
Contact Person Name: Shesagiri Rao
Contact Number: 9812312398
Job Title: AGM
Email: shesha@hal-india.com

Address for Communication
Company Name: SLRDC
Address: HAL, Hyderabad-500042 (Telangana)
Post Office: Balanagar

Organisation Approval Details
DOA Details:
DOA Scope:
POA Validity:
DOA Validity:
DOA Details:
POA Scope:

Air System Description
Air System Description: DAMS
Project Part Number: N146789
Project Details: digital audio management system
IMTAR Sub-Parts: C1 - Airbone Store Leading to Issue of Type Approval

Product Breakdown Structure
Product Breakdown Structure: System Safety Assessment.pdf

Edit **Submit** **Cancel**

Date: 26-06-2023

Figure 18 Project Registration Form: View Application

Registered Projects
New Project Registration
Help
Logout

Design Agency ID - DA0000000002
Design Agency Name - SLRDC
Date: 26-Jun-2023

Project Registration Form
Success
Registration request submitted successfully
Download Registration Form

Applicant Reference
Applicant Reference: DA_000000000226062023215714

Contact Person
Contact Person Name: Shesagiri Rao
Contact Number: 9812312398
Job Title: AGM
Email: shesha@hal-india.com

Address for Communication
Company Name: SLRDC
Address: HAL, Hyderabad-500042 (Telangana)
Post Office: Balanagar

Organisation Approval Details
DOA Details:
DOA Scope:
POA Validity:
DOA Validity:
DOA Details:
POA Scope:

Air System Description
Air System Description: DAMS
Project Part Number: N146789
Project Details: digital audio management system
IMTAR Sub-Parts: C1 - Airbone Store Leading to Issue of Type Approval

Figure 19 Submitting the form

2.4 Help

User can click on the help tab to view the user manual as shown in Figure-21.



Figure 20 Help file

2.5 Logout

After completion of the document/file transactions user may opt to logout. During logout if observations sent by CEMILAC are not downloaded/saved, the prompt for downloading the file as shown in following Figure-22 is displayed and **Logout** is not allowed. After saving all the observations the Logout is allowed and logout page is displayed.

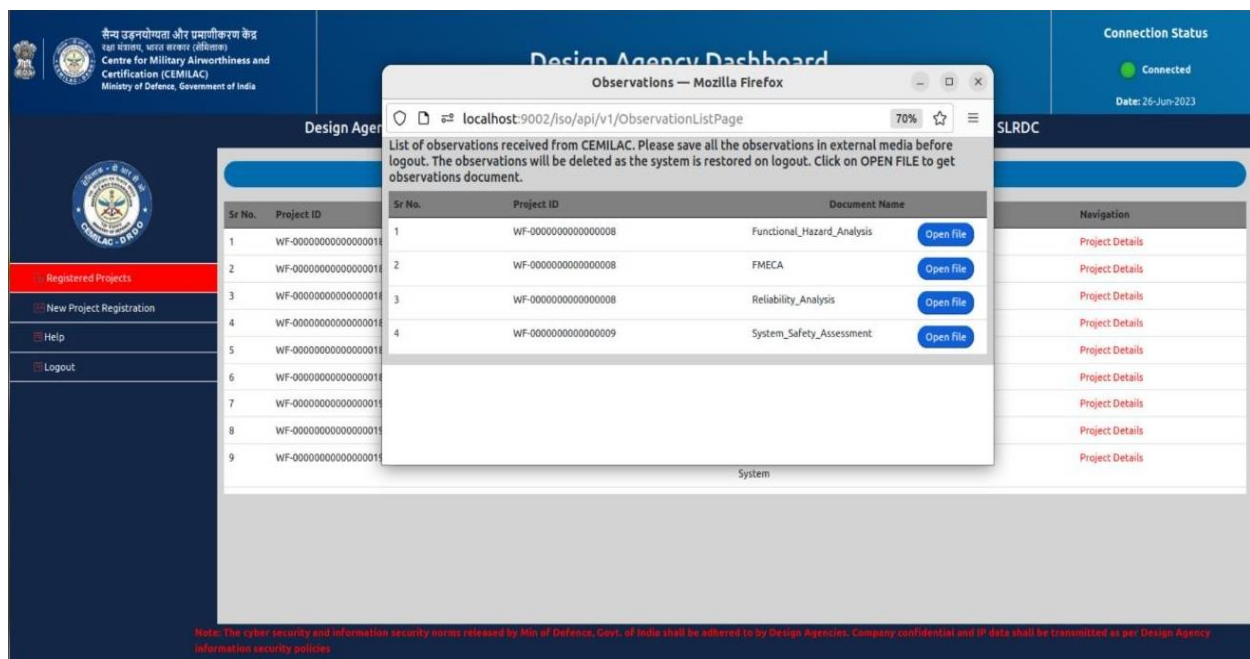


Figure 21 Prompt for observation download



Logout is successful. Please close the window.



Figure 22 Login Page after Logout

3 Error Code

Error codes related with Files upload			
Sl. No.	CAIR Code	Error code	Reason
1.	-1	101	File to be uploaded doesn't exist
2.	-2	102	socket() failed, unable to create socket
3.	-3	103	bind() failed
4.	-4	104	connect() failed, server not ready
5.	-5	105	recv() failed, no data received from server
6.	-6	106	ACK not received from Server
7.	-7	107	Upload failed, final ACK not received from Server
8.	-10	110	only few files downloaded
9.	-13	113	Permission Denied to Upload files
10.	-16	116	send() failed
11.	-17	117	Dongle Error
12.	-18	118	Invalid Upload Mode

13.	-19	119	Transform failed
14.	-20	120	General Error
15.	-21	121	Auth/Nonce fail
16.	-23	123	Lock Unavailable
Error codes related with Files Download			
17.	-2	202	socket() failed, unable to create socket
18.	-3	203	bind() failed
19.	-4	204	connect() failed, server not ready
20.	-5	205	recv() failed, no data received from server
21.	-6	206	ACK not received from Server
22.	-8	208	File doesn't exist at server
23.	-9	209	No write permission in destination path
24.	-10	210	Only few files downloaded
25.	-11	211	Download failed
26.	-14	214	Permission Denied to Download files
27.	-16	216	send() failed
28.	-17	217	Dongle Error
29.	-18	218	Invalid Download Mode
30.	-19	219	De Transform failed
31.	-20	220	General Error
32.	-21	221	Auth/Nonce fail
33.	-23	223	Lock Unavailable
Error Code related with Static and Dynamic File Structure			
34.			301 Static file got Corrupt
35.			302 Dynamic file got Corrupt
Error Codes for Connection Fail			
36.	-2		402 socket() failed, unable to create socket
37.	-3		-bind() failed
38.	-12		412 Server not reachable
39.	-17		417 Dongle Error